VINITH KUMAR T

Full-Stack Robotics & Autonomous Systems Engineer

Dharmapuri, Tamil Nadu, India | +91 XXXXX XXXXX | robologautomation@gmail.com | linkedin.com/in/vinith-kumar-t | github.com/vinithkumar-Themorgan | vinithkumar-themorgan.github.io

SUMMARY

Full-Stack Robotics Engineer with 2+ years building autonomous systems end to end — mechanical design & fabrication, embedded electronics, ROS/ROS2 autonomy, computer vision, and RF. National hackathon winner (KAVACH 2023), patent applicant, and published author, delivering AMRs, drones, six-axis manipulators and RF hardware from concept to reliable deployment.

EXPERIENCE

Robotics Engineer — Full-Stack Robotics & Autonomous Systems, Robolog Automation — Jun 2024 – Present · Dharmapuri, TN

- Build autonomous systems end to end — mechanical, electronics, embedded firmware, ROS/ROS2 autonomy, computer vision, and operator dashboards.
- Developed two EdTech AMRs with full navigation stacks (one ROS, one ROS2); these platforms helped the company secure a ₹1-crore project.
- Built NETSCOUT — an autonomous network-scouting AMR: WiFi-analysis turret, custom waypoint control (5–10 mm accuracy), and WiFi heatmapping over a LiDAR map.

Robotics Engineer — R&D (Contract), Bannari Amman Institute of Technology — Mar 2024 – May 2024 · Sathyamangalam, TN

- Designed and developed an Autonomous Mobile Robot (ROS) for Intec Expo 2024 — owned the navigation stack, sensor interfacing, and real-time control.
- Implemented path planning and obstacle avoidance; integrated hardware and software for a reliable live demonstration.

KEY PROJECTS

Autonomous Six-Axis Sorting Robot PATENT APPLIED

From-scratch 6-DOF arm with custom inverse kinematics and deep-learning 3D vision for autonomous warehouse package sorting. (Flipkart GRID 5.0)

Next-Gen Crime Detection System KAVACH 2023 — WINNER

Deep-learning vision on CCTV that flags criminal activity in real time to a centralized incident dashboard. Featured in The Hindu.

mmWave Waveguide — Single-Person Vital-Sign Detection R&D

Custom-fabricated pyramidal waveguide focusing a 60 GHz radar beam to isolate one subject for accurate heart-rate & respiration sensing.

Medical Emergency Response Drone SIH 2022 — 2ND RUNNER-UP

GPS-guided drone that reaches accident sites and triages injury severity via deep learning, alerting the nearest hospital.

ACHIEVEMENTS & RECOGNITION

- **KAVACH 2023** — National Winner (Govt. of India)
- **Smart India Hackathon 2022** — 2nd Runner-Up (₹50,000)
- **Flipkart GRID 4.0** — Excellence Award (Robotics)
- **Flipkart GRID 5.0** — National Finalist
- **Hackfest 2023** — Finalist (Mistral internship offer)
- **Aerathon 2023 (SAE)** — 23rd, national
- **Patent Applied** — Six-Axis Pick-&-Place Robot (3D camera)
- **Publication** — Quadruped robot, IRE Journals 7(4), 2023

TECHNICAL SKILLS

Robotics & Autonomy	ROS / ROS2, SLAM, Nav2, Perception, Forward & Inverse Kinematics, Sensor Fusion, Foxglove Studio
Computer Vision / AI	OpenCV, YOLO, Roboflow, Deep Learning, Object Detection
Programming	Python, C++
Embedded & Electronics	ESP32 / STM32, UART · I^{2}C · SPI · CAN · RS-485, sensor / driver / encoder integration, testing & troubleshooting
Drones & RF	Drone design, Pixhawk, PX4 / ArduPilot, CFD simulation; Waveguide design, RF analysis, mmWave
CAD, Mechanical & Fabrication	SolidWorks (+ simulation), Onshape, sheet metal, gears & transmissions, 3D printing, MIG welding, laser/CNC, lathe & milling
IoT, HMI & Controls	IoT (MQTT), Node-RED, HMI design, control-panel wiring

EDUCATION

B.E. — Electronics & Instrumentation (Robotics & Automation), Bannari Amman Institute of Technology — Oct 2020 – May 2024 · CGPA 8.1